A Modern Handbook of Physics

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Preface

The basic sciences and physics, in particular, are of prime importance today in the training of engineers for the various branches of the national economy. This has led to fundamental changes in recent years in the teaching of physics in engineer ing institutes, and to the students of other educational institutions in which they do not major in physics. The scope and scientific level of physics courses have been substantially supplemented and cover the main trends in the development of modern physics. Consequently, the physics textbooks for engineering students have inevitably become three-volume editions of almost fifteen hundred pages. The need has arisen, in this connection, for a concise handbook on this subject. The aim of the authors was to fulfill this need.

In scope and depth this handbook includes all of the definitions, formulas and information covered in the most comprehensive and up-to-date physics courses of engineering institutes and the physics departments of universities and colleges. Physical laws are concisely formulated, all the necessary explanations are given and, in many cases, derivations as well. Though it plays a vi-

tal role in the teaching of physics, experimental material has been omitted. This is due only to a lack of space. All the units and symbols comply with the requirements of the SI. Units of physical quantities and systems of units are listed and dealt with in a short appendix.

This handbook is designed primarily for engineering students, as well as college and university students studying, but not majoring in physics. It can be used to advantage by engineers and graduate students, as well as by instructors and lecturers of intermediate schools and colleges.

Mathematical knowledge required in using the handbook is within the scope of the ordinary mathematics courses of engineering institutes. The detailed index and the numerous cross references, indicating the chapter, section and subsection, are of aid in finding any required information. The authors are sincerely grateful to the translator, N. Weinstein, and the science editor, T. Baranovskaya, for their comprehensive and expert work in the publication of the English edition of this handbook, as well as for their valuable suggestions, which should facilitate its use by the reader.

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Parts One, Four, Five and the appendices were written by Prof. A. A. Detlaf; Parts Two, Three, Six, Seven and Eight, by Prof. B. M. Yavorsky.

Part I

Mechanics

1 Kinematics

1.1 Mechanical Motion. The Subject Matter of Mechanics

1.1.1

The simplest form of motion in nature is *mechanical motion*. It consists of changes in the relative positions of bodies or their parts in space in the course of time. The branch of physics dealing with the laws of mechanical motion is called *mechanics*. In a more narrow sense of the word, this is often under stood to mean *classical mechanics*, which deals with the motion of macroscopic bodies travelling at velocities a great many times less than that of light in free space. Classical mechanics is based on Newton's laws and is therefore frequently referred to as *Newtonian mechanics*. The laws of motion of bodies at velocities close to that of light in free space are taken up in *relativistic mechanics* (**??**), and the laws of motion of micro particles (for example, the electrons in atoms, molecules, crystals, etc.) are the subject matter of *quantum mechanics* (**??**).

1.1.2

Classical mechanics comprises three basic branches: statics, kinematics and dynamics. Statics deals with the laws for the composition of forces and the conditions for the equilibrium of bodies. *Kinematics* provides a mathematical description of all possible kinds of mechanical motion, regardless of what causes each particular kind of motion. *Dynamics* investigates the effect of the interaction between bodies on their mechanical motion.

1.1.3

The mechanical properties of bodies are determined by their chemical nature, internal structure and state. These are beyond the scope of mechanics and are studied in other branches of physics. Hence, in accordance with the conditions of each specific problem on hand, various simplified models are employed to describe real bodies in motion. These include the particle, the perfectly rigid body, the perfectly elastic body, the perfectly inelastic body and others.

A *material point*, or *particle*, is a body whose shape and size are of no consequence under the conditions of the problem being considered. For example, in a first approximation, the motion of a ship from one point to another can be dealt with as the motion of a particle. If, however, it becomes necessary to take into account such details of this motion as the pitching and rolling of the ship by swells or heavy seas, the ship should be dealt with as an extended body of definite shape. In the literature the word particle is more frequently used than material point.

Any extended body or system of such bodies, making up the *mechanical system* being investigated, can be regarded as a *system of particles*. For this purpose, ail the bodies of the system can be conceived of as being broken down into such a large number of parts that the size of each part is negligibly small compared to the sizes of the bodies themselves.

1.1.4

A *perfectly rigid body*, or simply *rigid body*, is one whose deformation can be neglected under the conditions of the given problem. The distance between any two points of a rigid body remains constant no matter what action the body is subjected to. A perfectly rigid body can be regarded as a system of particles rigidly connected to one another.

A *perfectly elastic body* is one whose deformation obeys Hooke's law (**??**). After the external forces applied to such a body have been removed, its initial size and shape are completely restored.

A *perfectly inelastic body* is one that completely retains its deformed state, caused by external forces applied to the body, when these forces have been removed.

1.2 Frames of Reference. Path, Path Length and Displacement Vector of a Particle

1.2.1

The position of a body in space can be specified only in reference to other bodies. It makes sense, for instance, to speak of the location of the planets with respect to the sun, that of an air liner or steamship with respect to the earth, but we cannot indicate their position in space "in general" without reference to some specific body. A perfectly rigid body in which a coordinate system is rigidly fixed, furnished with a "clock" and employed to determine the position of bodies and particles being investigated in space at various instants of time, is called a frame of reference. Sometimes the frame of reference is the coordinate system itself if it is provided with a "clock" and the solid in which the system is rigidly fixed is called the body of reference. In each specific problem the frame of reference is selected so as to simplify the problem to the maximum extent. Inertial frames of reference (**??**) are the most commonly used in physics.

1.2.2

Most extensively employed is the Cartesian rectangular system of coordinates (Figure 1.1) whose orthonormal (normal orthogonal) basis is formed by the three vectors **i**, **j** and **k**, which are unit vectors with respect to their magnitudes, mutually orthogonal and pass through the origin O of coordinates. The position of an arbitrary point M is specified by the radius vector **r**, which connects the origin O of coordinates to point M. Vector **r** can be resolved along the axes **i**, **j** and **k** of the basis as follows:

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

where $x\mathbf{i}$, $y\mathbf{j}$ and $z\mathbf{k}$ are the components of vector $\mathbf{r}$ along the coordinate axes. The coefficients of resolution, x, y and z, are the Cartesian coordinates of point M and, due to the orthogonality of the vectors of the basis, the projections of radius vector $\mathbf{r}$ on the corresponding coordinate axes.

The motion of a particle is completely specified if three continuous and single-valued functions of the time t,

$$x = x(t),\ y = y(t) \text{ and } z = z(t),$$

are given to describe the changes in the coordinates of the particle with time.

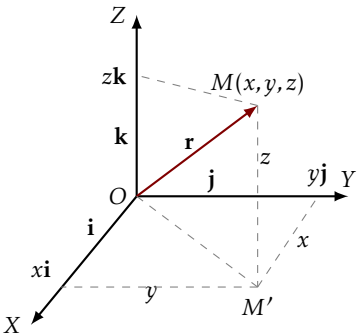


Figure 1.1: Cartesian system of coordinates.

These equations are said to be the *kinematic equations of motion of a particle*. They are equivalent to the single vector equation $\mathbf{r} = \mathbf{r}(t)$, for a moving particle.

1.2.3

The line described by a moving particle in space is called its *path*. The kinematic equations of motion of a particle specify the equation of its path in parametric form (the parameter being the time t). Depending upon the shape of the path, distinction is made between *rectilinear* and

curvilinear motion. The motion of a particle is said to be *plane* if its path lies completely in a single plane.

The mechanical motion of a body is relative, i.e. the nature of the motion and, in particular, the kind of path travelled by the points of the body depend upon the chosen frame of reference.

1.2.4

In the general case, the path of a moving particle is a spatial, rather than plane, curve. Introduced for such curves is the concept of an *osculating plane*. The osculating plane at the arbitrary point M of a curve is in the limiting position of a plane passing through any three points of the curve when these points approach point M without limit.

An *osculating circle* at point M of a curve is the limit of a circle passing through three points of the curve being considered when these points approach point M without limit. The osculating circle lies in the osculating plane. The centre of the osculating circle and its radius are called the *centre of curvature* and *radius of curvature*, at point M, of the curve being considered. The straight line connecting point M to the centre of curvature is called the *principal normal* to the curve at point M. The tangent to the curve at point M is perpendicular to the principal normal and also lies in the osculating plane.

1.2.5

The *path length* is the sum of the lengths of all the portions of the path over which the particle passes m the given interval of time. The instant of time $t = t_0$, from which we begin to consider the motion of a particle, is called the initial instant of time, and the position of the particle at this instant (point A in Figure 1.2) is called its initial position. Owing to the arbitrary nature of the time reference point it is usually assumed that $t_0 = 0$.

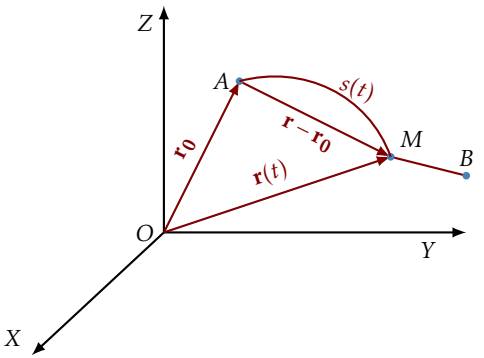


Figure 1.2: A particle moving on a curve.

The path length s, travelled by the particle from its initial position, is a scalar function of time: $s = s(t)$. As is obvious from its definition, the path length of a particle cannot be a negative quantity. If the particle travels along an arc of the path AB (Figure 1.2), always in the same direction and is at point M at the instant of time t,then $s(t) = \widehat{AM}$. If the particle travels along the path in a more complicated way, for example if it trav-

els from A to B, reaching B at the instant of time $t_1 < t$, and then reverses, returning to point M at instant t, then $s(t) = \widehat{AB} + \widehat{BM}$.

1.2.6

The displacement vector of a moving particle during the time interval from $t = t_1$ to $t = t_2$ is the vector from the, position of the particle at the instant of time t_1 to its position at instant t_2. It is equal to the increment in the radius vector of the particle during the time interval being considered:

$$\mathbf{r_2} - \mathbf{r_1} = \mathbf{r}(t_2) - \mathbf{r}t_1$$

The displacement vector is always directed along the chord subtending the corresponding portion of the path.

Shown in Figure 1.2 is the displacement vector of the particle for the time interval from t_0 to t. It is equal to $\mathbf{r} - \mathbf{r_0} = \mathbf{r}(t) - \mathbf{r}(t_0)$.

The displacement vector of a moving particle for the time interval from t to $t + \Delta t$ equals

$$\Delta r = r(t + \Delta t) - r(t) = \Delta x \times \mathbf{i} + \Delta y \times \mathbf{j} + \Delta z \times \mathbf{k}$$

where Δx, Δy and Δz are the increments (changes) in the coordinates of the particle during the time interval being considered.

1.2.7

A particle travelling freely in space can accomplish only three *independent motions*, i.e. motions that cannot be

represented as combinations of the other two. As a matter of fact, the motion of a particle along each of the axes of a Cartesian rectangular coordinate system cannot be obtained by its motion along the other two axes. The number of independent motions that a mechanical system can accomplish is called the *number of degrees of freedom* of the system. Hence, a free particle has three degrees of freedom.

1.3 Velocity

1.3.1

The concept of velocity is introduced in mechanics to characterize the rate of motion of bodies. It is the time rate of change in the position of moving bodies. The *average velocity* of a moving particle in the time interval from t to $t+\Delta t$ is the vector $\mathbf{v_{av}}$, equal to the ratio of the increment $\Delta\mathbf{r}$ in the radius vector of the particle during this time interval to the length of the time interval Δt:

$$\mathbf{v_{av}} = \frac{\Delta\mathbf{r}}{\Delta t}$$

Vector $\mathbf{v_{av}}$ has the same direction as $\Delta\mathbf{r}$, i.e. along the chord subtending the corresponding portion of the particle's path.

1.3.2

The *velocity* (or *instantaneous velocity*) of a moving particle is the vector quantity $\mathbf{v}$, equal to the first time deriva-

tive of the radius vector $\mathbf{r}$ of the particle being considered:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$

The velocity of the particle at the instant of time t is the limiting value that the average velocity $\mathbf{v_{av}}$ approaches for an infinitely small value of the time interval Δt. Thus

$$\mathbf{v} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{v_{av}}$$

The velocity vector $\mathbf{v}$ of the particle has the direction along a tangent to the path toward the motion in the same way as the vector $d\mathbf{r} = \mathbf{v}dt$ of infinitely small displacement of the particle during an infinitely short time interval dt.

The path length ds travelled by the particle during the length of time dt is equal to the magnitude of the displacement vector: $ds = |d\mathbf{r}|$. Therefore the magnitude of the velocity vector of a moving particle is equal to the first time derivative of the path length:

$$v = |\mathbf{v}| = \frac{\mathrm{d}s}{\mathrm{d}t}$$

1.3.3

The resolution of vector $\mathbf{v}$ along the basis of a Cartesian rectangular coordinate system is of the following form:

$$\mathbf{v} = v_x \times \mathbf{i} + v_y \times \mathbf{j} + v_z \times \mathbf{k}$$

The projections of the velocity of a moving particle on the coordinate axes are equal to the first time derivative

of the corresponding coordinates of the particle:

$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t},\ v_y = \frac{\mathrm{d}y}{\mathrm{d}t},\ v_z = \frac{\mathrm{d}z}{\mathrm{d}t}$$

and the magnitude of the velocity vector, sometimes called the *speed*, is

$$v = |\mathbf{v}| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2}$$

1.3.4

In rectilinear motion of a particle, the direction of its velocity vector remains constant. The motion of a particle is said to be *uniform* if the magnitude of its velocity does not change with time: $v = ds/dt = \text{const}$. The path length s travelled by a uniformly moving particle is a linear function of time: $s = vt$ (provided that $t_0 = 0$, see 1.2.5).

If the magnitude of the particle's velocity, i.e. its speed, increases with time ($dv/dt > 0$), the motion is said to be *accelerated*; if the speed decreases with time ($dv/dt < 0$), the motion is said to be *decelerated*.

1.3.5

The *average velocity of nonuniform motion* of a particle over a given portion of its path is the scalar v_{av}, called the *travelling speed*, which is equal to the ratio of the length Δs of this portion to the length Δt of the time interval required to travel over this portion:

$$v_{\text{av}} = \frac{\Delta s}{\Delta t}$$

It is equal to the speed of uniform motion in which the time required to travel the same path length As is the same as in the nonuniform motion being considered.

In curvilinear motion of a particle $|\Delta\mathbf{r}| < \Delta s$. Therefore, in the general case, the average velocity (travelling speed) v_{av} of the particle is not equal to the magnitude of its average velocity $\mathbf{v_{av}}$ over the same portion of its path (1.3.1): $v_{\text{av}} \geqslant |\mathbf{v}|$ where the equality sign corresponds to a rectilinear portion of the path.

1.3.6

In the case of plane motion of a particle M (1.2.3), it often proves convenient to use the polar coordinates r and φ, where r is the distance from pole O to particle M and φ is the vectorial angle measured from polar axis OA (Figure 1.3). The velocity $\mathbf{v}$ of particle M can be resolved

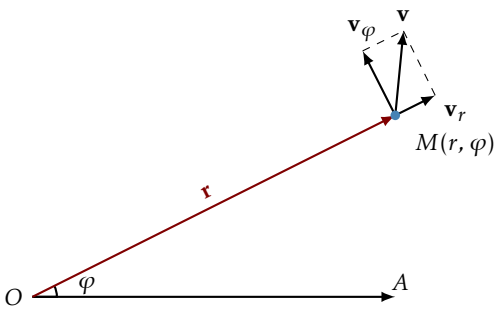


Figure 1.3: Polar coordinates of a moving particle.

into two mutually perpendicular components: the *radial*

velocity $\mathbf{v}_r$ and the *transverse velocity* $\mathbf{v}_\varphi$:

$$\mathbf{v} = \mathbf{v}_r + \mathbf{v}_\varphi$$

in which

$$\mathbf{v}_r = \frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{r}, \text{ and } \mathbf{v}_\varphi = \frac{\mathrm{d}\varphi}{\mathrm{d}t}[\mathbf{kr}]$$

where $\mathbf{r}$ is the polar radius vector of particle M, and $\mathbf{k}$ is the unit vector directed perpendicular to the plane of motion of the particle so that the rotation of radius vector $\mathbf{r}$ upon an increase in angle φ is seen, from the head of unit vector $\mathbf{k}$, to be counterclockwise.

The magnitude v of the velocity vector of particle M, in plane motion, is

$$v = \sqrt{\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(r\frac{\mathrm{d}\varphi}{\mathrm{d}t}\right)^2}$$

During the infinitely short length of time dt, polar radius vector $\mathbf{r}$ of the particle in plane motion sweeps over a sector of a circle, the area of the sector being $dS = r^2\,d\varphi/2$. Hence, the quantity

$$\sigma = \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{2}r^2\frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{1}{2}rv_\varphi$$

is called the *areal* velocity.

1.4 Acceleration

1.4.1

The concept of acceleration is introduced in mechanics to characterize the rate of change of the velocity vector

of a mov ing particle. The *average acceleration* in the time interval from t to $t + \Delta t$ is the vector $\mathbf{a}_{av}$, which is equal to the ratio of the increment $\Delta\mathbf{v}$ of the velocity vector of the particle, during the lime interval, to the length of the time interval Δt:

$$\mathbf{v}_{av} = \frac{\Delta \mathbf{v}}{\Delta t}$$

1.4.2

Acceleration (or *instantaneous acceleration*) of a moving particle is the vector quantity **a**, equal to the first time derivative of the velocity **v** of the particle being considered or, other wise, the second time derivative of the particle's radius vector **r**:

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

The acceleration of a moving particle at the instant of time t is the limiting value that the average acceleration $\mathbf{a}_{av}$ approaches for an infinitely small time interval Δt. Thus

$$\mathbf{a} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{v}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{a}_{av}$$

1.4.3

The resolution of vector **a** along the basis of a rectangular Cartesian coordinate system is of the form:

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

The projections of the acceleration vector of a moving particle on the coordinate axes are equal to the first time

derivatives of the corresponding velocity projections or, otherwise, the second time derivatives of the corresponding coordinates of the particle:

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2},\ a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2},\ a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}$$

The magnitude of the acceleration vector is

$$a = |a| = \sqrt{\left(\frac{dv_x}{dt}\right)^2 + \left(\frac{dv_y}{dt}\right)^2 + \left(\frac{dv_z}{dt}\right)^2}$$
$$= \sqrt{\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2 + \left(\frac{d^2z}{dt^2}\right)^2}$$

1.4.4

The acceleration vector of a moving particle lies in the osculating plane (1.2.4) passing through point M of the path being considered and is direct ed toward the concavity of the path BC (Figure 1.4). In this plane the acceleration vector $\mathbf{a}$ can be resolved into two mutually perpendicular components $\mathbf{a}_\tau$ and $\mathbf{a}_n$:

$$\mathbf{a} = \mathbf{a}_\tau + \mathbf{a}_n$$

1.4.5

Component $\mathbf{a}_\tau$ is called the *tangential acceleration* of the moving particle. It is directed along the tangent to the path and equals

$$\mathbf{a}_\tau = \frac{dv}{dt}\tau \text{ and } a_t = \frac{dv}{dt}$$

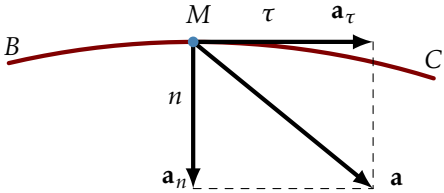


Figure 1.4: Tangential and normal components of the acceleration of a moving particle.

where $\tau = \mathbf{v}/v$ is the unit vector of the tangent, from point M of the path in the direction of velocity $\mathbf{v}$ of the particle, and a_τ is the projection of the tangential acceleration on the direction of vector $\mathbf{v}$. The tangential acceleration characterizes the rate of change of the magnitude of the particle's velocity vector. Vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ coincide in direction, i.e. $a_\tau > 0$, in accelerated motion (1.3.4); vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ are opposite in direction $a_\tau < 0$, in decelerated motion of the particle; and $a_\tau = 0$ in its uniform motion. If $a_\tau = \text{const} \neq 0$, the motion is said to be uniformly variable. In uniformly variable motion the magnitude of the particle's velocity is linearly time-dependent:

$$v = v_0 + a_\tau t$$

where $v_0 = v(0)$ is the magnitude of the initial velocity, i.e. the velocity at the initial instant of time $t = 0$. If $a_\tau = \text{const} > 0$, the motion of the particle is said to be *uniformly accelerated*, whereas if $a_\tau = \text{const} < 0$, it is said to be *uniformly decelerated*.

1.4.6

Component $\mathbf{a}_n$ of acceleration $\mathbf{a}$ of the moving particle is called its *normal acceleration*. It is directed along the principal normal to the path at point M being considered, toward the centre of curvature (1.2.4) of the path. Therefore, an is often called the *centripetal acceleration* of the particle. The normal acceleration equals

$$\mathbf{a}_n = \frac{v^2}{R}\mathbf{n}$$

where $\mathbf{n}$ is the unit vector of the principal normal, and R is the radius of curvature of the path. The normal acceleration characterizes the rate of change in direction of the particle's velocity vector.

If the particle travels in a straight line the normal acceleration $\mathbf{a}_n = 0$ and the acceleration is equal to the tangential acceleration of the particle: $\mathbf{a} = \mathbf{a}_\tau$.